Combustion and Flame 159 (2012) 3235–3241

Contents lists available at SciVerse ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Shock tube studies of methyl butanoate pyrolysis with relevance to biodiesel

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ARTICLE INFO

Article history:
Received 12 May 2012
Received in revised form 14 May 2012
Accepted 16 May 2012
Available online 28 June 2012

Keywords:
Methyl butanoate
Biodiesel
Kinetics
Pyrolysis
Shock tube

ABSTRACT

Methyl butanoate pyrolysis and decomposition pathways were studied in detail by measuring concentration time-histories of CO, CO_2, CH_3, and C_2H_4 using shock tube/laser absorption methods. Experiments were conducted behind reflected shock waves at temperatures of 1200–1800 K and pressures near 1.5 atm using mixtures of 0.1%, 0.5%, and 1% methyl butanoate in Argon. A novel laser diagnostic was developed to measure CO in the v_1 fundamental vibrational band near 4.56 μm using a new generation of quantum-cascade lasers. Wavelength modulation spectroscopy with second-harmonic detection (WMS-2*f*) was used to measure CO_2 near 2752 nm. Methyl radical was measured using UV laser absorption near 216 nm, and ethylene was monitored using IR gas laser absorption near 10.53 μm. An accurate methyl butanoate model is critical in the development of mechanisms for larger methyl esters, and the measured time-histories provide kinetic targets and strong constraints for the refinement of the methyl butanoate reaction mechanism. Measured CO mole fractions reach plateau values that are the same as the initial fuel mole fraction at temperatures higher than 1500 K over the maximum measurement time of 2 ms or less. Two recent kinetic mechanisms are compared with the measured data and the possible reasons for this 1:1 ratio between MB and CO are discussed. Based on these discussions, it is expected that similar CO/fuel and CO_2/fuel ratios for biodiesel molecules, particularly saturated components of biodiesel, should occur.

1. Introduction

There is an environmentally-critical need to understand the combustion chemistry of carbon–neutral [1,2] sources of fuels such as biodiesel. Biodiesel fuel is normally in the form of long-chain esters and is produced by trans-esterification of long-chain fatty acids that are derived from oils, animal fats [3] or algae [4]. Biodiesel, while similar to conventional diesel in some respects, can have quite different combustion characteristics, reactivity, and pollutant formation due to the oxygenated chemical structure, and only limited high-temperature data are available for the reactivity of these complex molecules. Shock tube/laser absorption studies of methyl ester pyrolysis and oxidation can provide the needed species time-history targets to refine and validate detailed mechanisms for these fuels.

Biodiesel derived from rapeseed and soybean mainly contain five components: methyl palmitate ($C_{17}H_{34}O_2$), methyl stearate ($C_{19}H_{38}O_2$), methyl oleate ($C_{19}H_{36}O_2$), methyl linoleate ($C_{19}H_{34}O_2$), and methyl linolenate ($C_{19}H_{32}O_2$). There have been only a few studies [5–8] on the chemical kinetics of these large methyl esters because of their low vapor pressure and chemical complexity. Methyl butanoate (MB), $CH_3{-}CH_2{-}CH_2{-}C({=}O)O{-}CH_3$, has been used as a surrogate for long-chain methyl esters by several researchers because it contains much of the essential chemical structure of its long-chain counterparts, i.e., the methyl ester termination and a shorter, but similar alkyl chain, and because detailed reaction mechanisms describing this fuel are of a manageable size. Especially important, as shown in the current study, is the fact that methyl butanoate does exhibit some limiting kinetic characteristics that can be directly applied to longer chain esters and biodiesel.

Recent studies, however, have shown that there are some combustion-related differences in the behavior of methyl butanoate and biodiesel. Methyl butanoate is less reactive than soybean biodiesel [9], has a lower cetane number [10], and does not exhibit the negative temperature coefficient (NTC) behavior observed for larger methyl esters [11]. Understanding these differences is critical to properly implementing the hierarchical mechanism development strategy for larger methyl esters [12].

A number of studies have previously been carried out on the detailed chemical kinetics of methyl butanoate. The first detailed kinetic model for the oxidation of MB was developed by Fisher et al. [13] in 2000 and consisted of 279 species and 1259 reactions.

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http://dx.doi.org/10.1016/j.combustflame.2012.05.013

This model was revised by Metcalfe et al. [14] to reproduce shock tube experimental data. Gaïl et al. [15] developed a modified mechanism to compare jet-stirred reactor, opposed-flow diffusion flame, and flow reactor data. More recently, Dooley et al. [16] further modified the Metcalfe et al. MB mechanism to reproduce their shock tube and rapid compression machine data and some other data available in literature. Huynh and Violi [17] used ab initio techniques to create a sub-mechanism for the thermal decomposition of methyl butanoate, and their modified model was in closer agreement to the experimental work of Farooq et al. [18] who measured CO_2 yields for MB in a shock tube. Recently, Walton et al. [19] measured the intermediates formed during the ignition of MB in a rapid compression machine. Further speciation data is needed, particularly from shock tubes, for the pyrolysis and oxidation of methyl butanoate to improve the understanding and predictive capability of MB mechanisms.

Here, we present a detailed experimental study on the pyrolysis of methyl butanoate behind reflected shock waves using laser absorption methods to measure time-histories of four species: CO, CO_2, CH_3, and C_2H_4. A novel laser diagnostic is developed to measure CO in the fundamental vibrational band using a new generation of quantum-cascade lasers. The measurements are compared to the predictions of two kinetic mechanisms and the influence of MB decomposition and H-abstraction reactions on the species profiles is discussed. These measurements can provide strong constraints on the decomposition pathways, reaction rates, and overall detailed MB reaction mechanism, and offer analogous predictions for CO and CO_2 during the pyrolysis of real biodiesel.

2. Experimental details

2.1. Shock tube facility

Experiments were performed in the reflected shock region of a high-purity, stainless steel, helium-driven shock tube at Stanford University. The driven section is 10.5 m long and the driver section 3.7 m long, with an inner diameter of 15.24 cm. This geometric configuration provided at least 2 ms of high-quality test time of uniform temperature and pressure for these studies; measurements were made at an axial location 2 cm from the end-wall. Test mixtures were prepared manometrically in a 40 l stainless steel tank heated uniformly to 50 °C, and mixed with a magnetically-driven stirring vane for at least 2 h prior to the experiments. Between experiments, the shock tube and mixing assembly were turbo-pumped to approximately 6 μtorr. Reflected shock temperatures and pressures were determined from the incident shock speed (extrapolated to the end-wall) using standard normal shock relations, with an uncertainty of ±0.7% and ±1%, respectively. Commercially available methyl butanoate (>99% pure) from Sigma–Aldrich and research grade (99.999%) Argon and Helium supplied by Praxair Inc. were used in the experiments.

2.2. Laser absorption measurements

2.2.1. Mid-IR absorption of CO

A new diagnostic was developed in this work for measuring CO concentration in the fundamental vibrational band of CO near 4.56 μm using a room-temperature quantum-cascade (QC) laser supplied by Alpes Lasers. The R(13) transition of the v_1 vibrational band of CO was chosen due to its relatively high absorption strength and negligible interference from CO_2 and H_2O. The room-temperature line-strength measured for this transition using a simple static cell setup was within 2% of the HITRAN [20] value ($6.0387\ cm^{-2}\ atm^{-1}$). Since Argon is the primary bath gas used in shock tube kinetic studies, the Argon-broadening parameters were measured for this transition behind reflected shock waves using scanned-wavelength direct absorption at a repetition rate of 10 kHz. Since the vibrational relaxation time of CO (in Ar) can be on the order of hundreds of microseconds over the temperature range of interest (1000–2000 K), the relaxation time was reduced to a few microseconds by adding 1% H_2 to the desired 0.5% CO/balance Ar mixture. The measured Argon-broadening coefficient ($2\gamma_{CO-Ar}$) over the temperature range of 1200–1900 K is 0.0762 ± 0.0008 cm^{-1}/atm with a temperature exponent of $n = 0.57$. The detection limit for this new diagnostic is approximately 5 ppm for most conditions. Quantitative CO concentration (mole fraction, X_{CO}) profiles are generated from the raw traces of fractional absorption using Beer's law, the measured line-strength (S), the path length L, and the measured line-shape function (ϕ):

$$I/I_0 = \exp(-S\phi P_{total} X_{CO} L) \tag{1}$$

2.2.2. IR diode laser absorption of CO_2

Absorption measurements of CO_2 were made using wavelength-modulation spectroscopy with second-harmonic detection (WMS-2*f*). A commercially available distributed feedback diode laser (from NanoPlus) emitting near 3633 cm^{-1} was used. This diagnostic was developed and demonstrated previously [21] for the measurement of CO_2 concentration and temperature in shock tube kinetic experiments. Line-strength, self-broadening and Ar-broadening coefficients were measured previously for the R(28) transition (at 2752.5 nm) used here [22]. Quantitative CO_2 concentration profiles were generated by comparing the measured 1*f*-normalized WMS-2*f* signals with simulated values. The WMS strategy was preferred over direct absorption to achieve higher signal-to-noise ratios in low fuel concentration experiments.

2.2.3. CO_2 laser absorption of C_2H_4

Ethylene mole fraction was measured by taking advantage of the fortuitous overlap of the P(14) line of the CO_2 gas laser at 10.532 μm with the strong ethylene absorption band near 10.6 μm. Experiments were also conducted at an off-line wavelength of 10.675 μm to correct for any interference absorption from fuel or other alkenes. The detection limit for ethylene measurements is approximately 100 ppm at a temperature of 1200 K and 2 atm. Experimental details of this diagnostic and high-temperature absorption cross-sections of ethylene and related alkenes are discussed in Ren et al. [23] and Pilla et al. [24].

2.2.4. Laser absorption of CH_3

Measurements of methyl mole fraction were made in the UV at 216.6 nm using the frequency-quadrupled output of near-infrared radiation from a pulsed Ti:Sapphire laser. Stable mode-locking of the Ti:Sapphire laser (MIRA HP, Coherent Inc.) was obtained at a wavelength of 866.4 nm with a peak output of 1 W at 76 MHz repetition rate. Deep UV light was generated by frequency conversion, using fourth-harmonic generation (FHG, Coherent Inc.), to obtain output of ~7 mW at 216.6 nm. To account for interference absorption from ethylene and higher olefins, measurements were also performed at an off-line wavelength of 218.7 nm. Using a common-mode-rejection scheme, a minimum absorbance of ~0.1% can be detected, resulting in a minimum detection of approximately 1 ppm at 1200 K and 2 atm. Further details of this laser setup can be found elsewhere [25].

3. Results

Representative data for simultaneous measurement of CO, CH_3, and C_2H_4 time-histories in a single reflected shock wave experiment are shown in Fig. 1 for an initial mixture of 0.5% MB in Argon, at an initial reflected shock temperature of 1486 K and pressure of 1.44 atm. Excellent signal-to-noise (SNR) ratio is achieved for all measured species over the wide range of concentrations and time scales. In the high fuel concentration experiments, the endothermic nature of pyrolysis can lower the test gas temperature significantly at long test times. For the 1% MB mixtures, this temperature drop ranges from about 10 K to 120 K for reflected shock temperatures of 1200–1600 K. To account for this effect, absorption cross-sections based on this lower temperature, calculated assuming the reactions occur at constant volume, were used to determine long-time species yields.

Measurements of CO_2 concentration were carried out in a 1% MB/Ar mixture to achieve desired high SNR and sensitivity. Fig. 2 shows simultaneous measurements of CO and CO_2 in a 1% MB/Ar mixture for initial reflected shock conditions of 1545 K and 1.35 atm. At 1500 μs, the measured mole fractions of CO and CO_2 are 9990 ppm and 4760 ppm, respectively. In this particular experiment, the CO and CO_2 measurements recover about 97% of the oxygen in the initial fuel. The oxygen balance was close to 99% in a similar experiment carried out at a reflected shock temperature of 1645 K. It should be noted that few, if any, of the O atoms ever get free of their C partners during the pyrolysis of MB and other esters. Tracking the oxygen-containing carbon molecules thus enables one to understand the fate of oxygen atoms and kinetically favored decomposition pathways in oxygenated fuels such as MB.

The complete set of CO measurements in 0.5% MB/Ar mixture are shown in Fig. 3 and indicate the extremely high sensitivity of the new CO diagnostic employed. A very intriguing aspect of these data is that at high temperatures the measured CO mole fraction asymptotes to 5000 ppm, the same value as the initial fuel concentration. Possible reasons for this apparent 1:1 ratio between MB and CO are discussed in Section 4.

A subset of the measured CO time-history data is compared with two MB kinetic mechanisms of Huynh and Violi [17] and Dooley et al. [16] in Fig. 4. The Huynh mechanism is closer to CO plateau levels at higher temperatures, though it does not accurately capture the initial CO formation and does not reach the 5000 ppm level over the 1 ms time. At lower temperatures, this mechanism significantly underestimates CO data over all times. The Dooley mechanism underestimates CO mole fraction at low temperatures and overestimates it at high temperatures. It should be noted that the Dooley mechanism was developed and validated using shock tube/RCM ignition data and speciation data from flow reactor, jet-stirred reactor and opposed-flow diffusion flame experiments. On the other hand, the success of the Huynh mechanism, in part, is because it has previously been validated against CO_2 time-history data obtained in a shock tube [18]. Sensitivity analyses carried out, using either mechanism, at various experimental conditions indicate that CO is strongly sensitive to the MB decomposition reaction:

$$MB \rightarrow CH_2(C{=}O)OCH_3(ME2J) + C2H5 \qquad (Rxn.1)$$

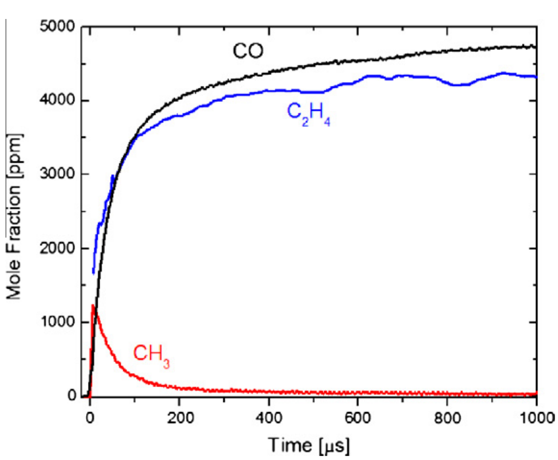

Fig. 1. Species time-history measurements for methyl butanoate (MB) pyrolysis. T_5 = 1486 K, P_5 = 1.44 atm, 0.5% MB in Ar.

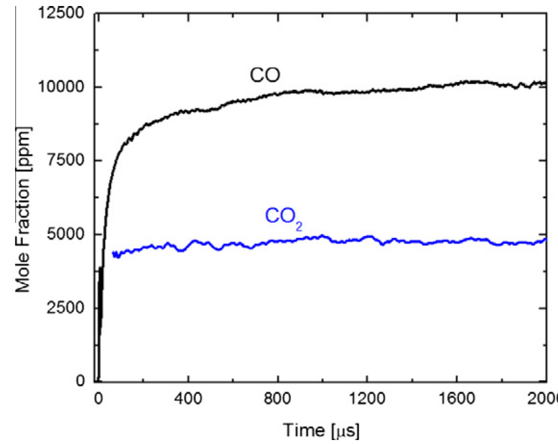

Fig. 2. Species time-history measurements for methyl butanoate (MB) pyrolysis. T_5 = 1545 K, P_5 = 1.35 atm, 1% MB in Ar.

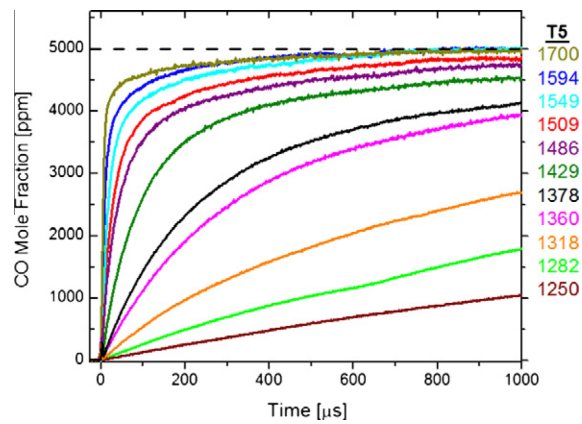

Fig. 3. CO time-history measurements for methyl butanoate (MB) pyrolysis. $P_5 \sim 1.5$ atm, 0.5% MB in Ar.

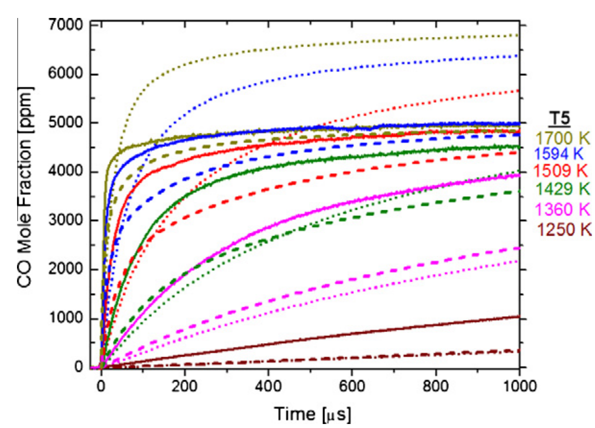

Fig. 4. Comparison of measured (solid lines) CO time-history data with Huynh et al. (dashed lines) and Dooley et al. (dotted lines). $P_5 \sim 1.5$ atm, 0.5% MB in Ar.

and the branching ratio of CH_3OCO

$$CH_3OCO \rightarrow CO_2 + CH_3 \quad \text{or } CH_3OCO \rightarrow CO + CH_3O \qquad \text{(Rxn.2)}$$

(Descriptor name of the intermediates, i.e. ME2J, are adapted from Fisher et al. [13].) The decomposition rate, while approximately same in the two models, is slow enough to show a disagreement between simulations and measurements at early times. The branching ratio of CH_3OCO favors CO formation in the Dooley mechanism and thus leads to overestimation of CO plateaus at high temperatures.

Time-history measurements of CO concentration were also carried out using 0.1% MB/Ar and 1% MB/Ar mixtures to study the effect of initial fuel concentration on CO plateau levels. The CO fractional yield, defined as the ratio of CO mole fraction to the initial fuel mole fraction, at 1 ms, is plotted in Fig. 5 as a function of temperature along with simulated fractional yields of the two chemical kinetic mechanisms considered. All measurements made with different initial fuel concentration plateau to a fractional yield of 1.0 for temperatures higher than 1500 K. Both mechanisms underpredict the CO yield by about 50% for $T < 1400$ K. At higher temperatures, the Huynh et al. model is closer to the measured yields than the Dooley et al. model. The measured yields indicate that for a given temperature the mixture with lower fuel concentration has slightly higher reactivity. The Dooley et al. model follows the same trend. However, the Huynh et al. model exhibits smaller reactivity at low fuel concentration. Measurements of CO time-history were also carried out at higher pressures (4–5 atm) but the fractional yields did not show any significant difference, indicating that the 1.5 atm experiments are close to the high-pressure limit.

A subset of the measured CH_3 time-histories are compared with the two kinetic mechanisms of Huynh et al. (dashed lines) and Dooley et al. (dotted lines) in Fig. 6. Sensitivity analyses carried out using either mechanism show that CH_3 peak is controlled by the fuel decomposition reaction:

$$MB \rightarrow CH_3CH_2CH_2(C{=}O)O(BAOJ) + CH_3 \qquad \text{(Rxn.3)}$$

whereas the removal rate is most sensitive to the methyl recombination reaction ($CH_3 + CH_3 \leftrightarrow C_2H_6$). The Dooley et al. model underestimates the CH_3 peak and removal rates at all temperatures. Similar to the CO data, the Huynh et al. model underpredicts CH_3 at lower temperatures but does a reasonable job at higher temperatures. While the methyl recombination rate is similar in the two models, the Huynh mechanism uses a faster rate for Rxn. 3 and thus predicts CH_3 peak values that are closer to measurements.

A subset of measured C_2H_4 time-history data are shown in Fig. 7 and compared with the kinetic mechanisms of Huynh et al.

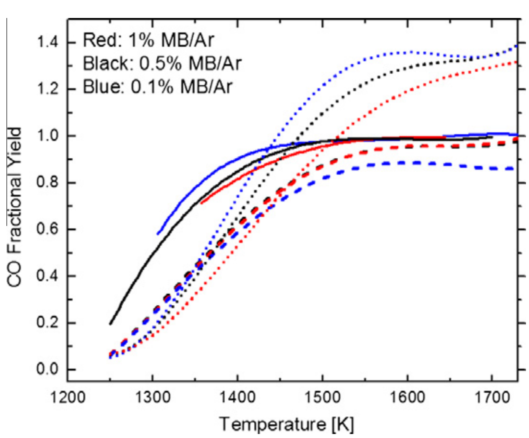

Fig. 5. Comparison of measured (solid lines) CO yields at 1 ms with simulated yields from Huynh et al. (dashed lines) and Dooley et al. (dotted lines). $P_5 \sim 1.3$–1.5 atm.

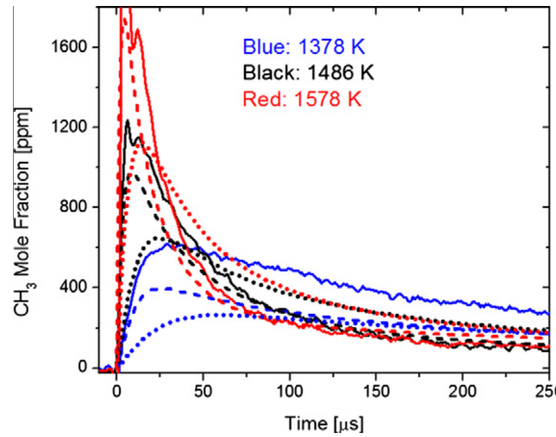

Fig. 6. Measured (solid lines) CH_3 time-histories plotted with simulations from Huynh et al. (dashed lines) and Dooley et al. (dotted lines). $P_5 \sim 1.5$ atm, 0.5% MB in Ar.

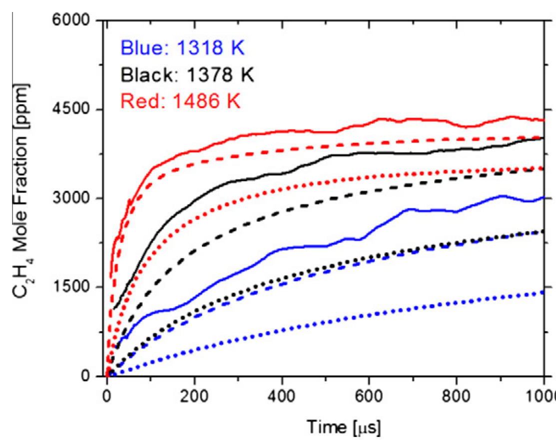

Fig. 7. Measured (solid lines) C_2H_4 time-histories plotted with simulations from Huynh et al. (dashed lines) and Dooley et al. (dotted lines). $P_5 \sim 1.5$ atm, 0.5% MB in Ar.

(dashed lines) and Dooley et al. (dotted lines). Both models underestimate the ethylene production, though the Huynh et al. model is closer to the measured data at higher temperatures. Sensitivity analyses show that C_2H_4 is sensitive to the two decomposition reactions listed above (Rxns. 1 and 3) and the branching ratio of n-C_3H_7. The higher rate used for Rxn. 3 in the Huynh model again results in better agreement with the measured ethylene profiles.

Fig. 8 shows simultaneous measurements of CO and CO_2 in 1% MB/Ar mixture at relatively low (1357 K) and high (1645 K) temperatures, compared with the Huynh et al. mechanism. At 1357 K, the model does an excellent job of predicting CO_2 data but falls well short of CO measurements. The model does a reasonable job of predicting both species at 1645 K but provides a CO/CO_2 oxygen balance of only 93% at 1000 μs versus the measured oxygen balance of approximately 99%.

4. Discussion

The modeling results using the Huynh and Violi mechanism [17] were examined to investigate the trends observed in the experiments and suggest generalizations to larger methyl ester fuels. Under the present pyrolysis conditions, the only radicals that contribute to fuel consumption are H atoms. Although MB contains O atoms, they are bonded tightly to C atoms within the MB

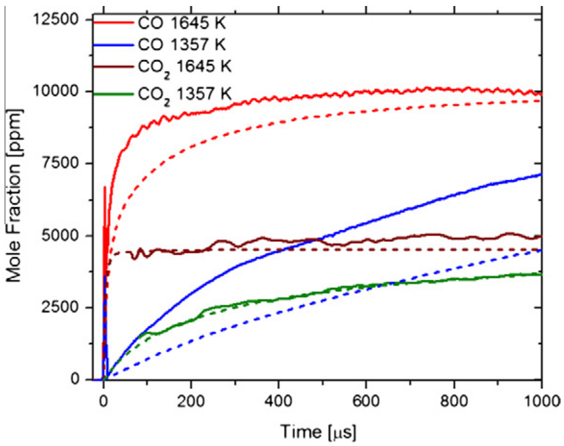

Fig. 8. Time-history measurements of CO and CO_2 along with simulations of Huynh et al. (dashed lines). $P_5 \sim 1.5$ atm, 1% MB in Ar.

molecule. The scission of the C═O/C—O bonds to produce isolated O atoms or OH radicals, even at temperatures as high as 1600 K, is not observed in either the experiments or the kinetic modeling results. Methyl (CH_3) does produce small amounts of methane by H atom abstraction from the MB fuel, but the methyl levels are always low enough such that those abstraction reactions contribute very little towards fuel consumption.

Sensitivity analyses show that the early time profiles of all measured species are sensitive to MB unimolecular decomposition reactions, particularly Rxns. 1 and 3 (Table 1). Increasing the rates of these two reactions in the Huynh model improves the agreement with measured data at lower temperatures. These reactions are more important at lower temperatures, while the H abstraction reactions (whose reaction rate constant estimates are likely to be more accurate) become more important at higher temperatures.

A key feature of the experiments is the ratio of production of CO and CO_2 and the intriguing observation that, within the precision of the experiments, one CO is produced for each MB consumed. The relative production of CO and CO_2 in methyl ester oxidation is quite important in practical terms, since O atoms included in diesel fuel have been shown to reduce soot production both experimentally [26] and in kinetic modeling [27]. However, O atoms that make CO_2 directly are less effective in soot reduction per O atom in the fuel than O atoms that produce CO. A thorough understanding of the relative production of CO and CO_2 can establish the soot reduction potential of biodiesel fuel and other oxygen-containing components as additives to conventional diesel fuel.

The kinetic model shows that MB consumption occurs initially via unimolecular decomposition, primarily those reactions breaking C—C or C—O bonds in MB. The C═O double bond appears to never be broken during MB reaction, and little initiation occurs via breaking C—H bonds in MB, due to the higher bond dissociation energy of C—H bonds. Initially at time zero (e.g., immediately behind the reflected shock wave in the experiment) at 1600 K and initial fuel mole fraction of 0.5% MB, the rates of the significant MB consumption reactions are summarized in Table 1 (from Huynh and Violi model [17]). The nomenclature for decomposition products is adopted from [13]. The products of these decompositions react further, leading to final products of the pyrolysis, and Table 1 shows the contributions of each decomposition reaction towards eventual production of CO or CO_2. In one case (Rxn.6), the production of $c_3h_7co + ch_3o$ splits MB into two products that each produce one CO molecule, and in another case (Rxn.1), the production of ME2J + C_2H_5, the subsequent β-scission decomposition of ME2J:

$$CH_2COOCH_3(ME2J) \rightarrow CH_2CO + CH_3O$$

also splits the two O atoms into species that each produce CO. The other three decomposition reactions of MB in Table 1 lead to CO_2 as a final product, never separating the two O atoms. Using the rates of the five reactions in Table 1, the decomposition reactions produce a ratio of CO/CO_2 approximately equal to 3/4. This ratio depends somewhat on the value of the initial reaction temperature, but it is consistently much smaller than the final overall pyrolysis product ratio measured to be approximately 2/1, as shown in Figs. 2 and 8.

As MB is consumed, H atom levels grow to a point where they can begin H atom abstraction reactions from MB that produce H_2. There are four possible abstractions possible in MB, shown in Table 2, with their rates at a time of 5 μs in the same 1600 K, 0.5% MB case used in Table 1. By this time, the MB mole fraction has decreased from 5000 ppm to about 1000 ppm, so the unimolecular decomposition rates, which are proportional to the concentration of MB, have fallen by a factor of 5 and are no longer the only reactions consuming MB. The H atom abstraction reactions can be seen to produce a ratio of CO/CO_2 of about 5/1, much larger than the ratio resulting from the decomposition reactions and much larger than the final measured ratio of 2/1. When the overall contributions of unimolecular decomposition and H atom abstraction are combined, the final ratio of CO produced to MB consumed is very close to 1/1 and the final ratio of CO/CO_2 is very close to 2/1, and the present mechanism shows that these final values are a complicated combination of the results of the decomposition over the entire pyrolysis history and the later-term results of the H atom abstraction reactions. The early-time decomposition of MB produces a slight excess of CO_2 over CO, and the H atom abstractions produce such a large excess of CO over CO_2 that the eventual ratios become 2/1.

Further examination of the computed results at other conditions of temperature and initial MB concentration finds that these competing processes always arrive at effectively the same final conditions. At temperatures as low as 1300 K, the decomposition reactions are all much slower than those at 1600 K, but at such lower temperatures the production of H atoms is much slower, so the onset of H atom abstraction reactions comes much later than at 1600 K, with the same overall ratio of products.

Two other smaller methyl esters have been studied under similar conditions in a concurrent study (not reported here). At high temperatures (>1500 K), measured CO mole fraction is 4200 ppm for 0.5% methyl acetate in Ar, and 4500 ppm for 0.5% methyl propionate in Ar, so the ratio of CO/fuel increases from 4200/5000 in methyl acetate to 4500/5000 for methyl propionate to 5000/5000 in methyl butanoate. The present model formulation can be used

Table 1
Unimolecular decomposition rates during 0.5% MB/Ar pyrolysis at 1600 K using Huynh model [17].

	Unimolecular decomposition	Reaction rate (mole/cm^3 s) at 1600 K, $t = 0$	Final products
Rxn. 1	MB = $CH_2(C{=}O)OCH_3$ (ME2 J) + C_2H_5	3.1×10^{-3}	CO + CO
Rxn. 3	MB = $CH_3CH_2CH_2(C{=}O)O$ (BAOJ) + CH_3	5.4×10^{-3}	CO_2
Rxn. 4	MB = C_3H_7 + CH_3OCO	2.5×10^{-3}	CO_2
Rxn. 5	MB = $CH_2CH_2(C{=}O)OCH_3$ (MP3J) + CH_3	2.4×10^{-3}	CO_2
Rxn. 6	MB = C_3H_7CO + CH_3O	0.7×10^{-3}	CO + CO

Table 2
H-atom abstraction rates during 0.5% MB/Ar pyrolysis at 1600 K using Huynh and Violi model [17].

	H-atom abstraction	Reaction rate (mole/cm^3 s) at 1600 K, 5 μs	Final products
Rxn. 7	MB + H=CH_3CHCH_2(C=O)OCH_3 (MB3J) + H_2	3.5×10^{-4}	CO_2
Rxn. 8	MB + H=$CH_3CH_2CH_2$(C=O)OCH_2 (MBMJ) + H_2	3.3×10^{-4}	CO + CO
Rxn. 9	MB + H=$CH_2CH_2CH_2$(C=O)OCH_3 (MB4J) + H_2	2.8×10^{-4}	CO + CO
Rxn. 10	MB + H=CH_3CH_2CH(C=O)OCH_3 (MB2J) + H_2	2.8×10^{-4}	CO + CO

to see how this ratio would change as the size of the methyl ester grows to the size of biodiesel molecules.

Larger methyl ester fuels can be thought of having all the same features as MB but with the addition of a large number of CH_2 groups into the hydrocarbon chain in MB. To consider biodiesel, one of the commonly found saturated methyl ester species is methyl stearate ($C_{19}H_{38}O_2$), with 15 "normal" CH_2 groups, one terminal CH_3 group, and one anomalous CH_2 group adjacent to the C=O in the ester group. In MB, there is only one "normal" CH_2 group, along with the same terminal CH_3 group and the anomalous CH_2 group adjacent to the C=O. These "normal" CH_2 groups in methyl stearate will have C—H bond energies that are essentially the same as the secondary C—H bonds in the same group in MB [28] and their H abstraction reaction rates will be the same as the abstraction rates at the MB3J site in MB, Rxn. 7 in Table 2.

Addition of CH_2 groups in the hydrocarbon chain provides more possibilities for the parent fuel to decompose and more sites for H atom abstraction by H atoms. For the decomposition reactions, this adds more internal C—C bonds to be broken, each of which produces a purely hydrocarbon alkyl radical and a methyl ester radical. The methyl ester radical has the structure of an alkyl radical with the methyl ester group on the end of the chain opposite to the radical site. Rxn. 1 in Table 1 has this form, with its ethyl radical and the ME2J methyl ester radical. New decomposition reactions in the case of methyl stearate have products of ME2J + $C_{16}H_{33}$, MP3J + $C_{15}H_{31}$, MB4J + $C_{14}H_{29}$ and others down to M16J + C_2H_5 and M17J + CH_3. Successive β-scissions of each of these methyl ester alkyl analog species will lead either to the ME2J radical or to the CH_3OCO radical, which decompose to produce CO + CO or CO_2, respectively, as seen in Rxns. 1 and 4 in Table 1. Each pair of CH_2 groups added to the hydrocarbon chain leads to two CO and one CO_2 species as eventual products.

For the H atom abstraction reactions, addition of more CH_2 groups leads to more examples of Rxn. 7 in Table 2, abstraction from a conventional secondary site in the carbon chain. The alkyl-like radicals produced in those reactions will decompose via a series of β-decomposition reactions, producing ethene molecules and a new alkyl-like radical shorter than the previous radical by 2 C atoms, eventually reaching either ME2J or CH_3OCO, just as noted for the decomposition reaction products. Each pair of CH_2 groups added to the hydrocarbon chain thus leads to two CO and one CO_2 products as results of H atom abstraction reactions.

This illustration shows that, for MB and all longer linear methyl esters, each addition of two CH_2 groups increases the average number of CO molecules by two, keeping pace with the lengthening of the fuel chain length, and one CO_2 molecule, keeping half the same pace. This rate of CO and CO_2 production exists for MB, and the same type of analysis shows that the same proportions of CO and CO_2 production should be expected for all linear methyl ester fuels up to and beyond the fuel size for biodiesel. A sample computation using the kinetic mechanism for the pyrolysis of methyl stearate [28] gives the same product ratio of CO equal to the initial concentration of the methyl stearate fuel with the CO_2 product level exactly half that value.

When methyl esters are burned, there are additional reaction pathways producing CO and CO_2, as well as the reaction CO + OH ↔ CO_2 + H, that are also very important, so the concept of a strictly pyrolytic process controlling MB decomposition is no longer correct. However, soot formation in diesel engines takes place under very fuel-rich conditions [10], so the present analysis is not irrelevant in such environments. In particular, knowledge of the reaction pathways for direct production of CO and CO_2 from the ester group in methyl ester fuels provides an insight into the magnitude of the soot reduction potential of oxygen atoms in biodiesel fuels. In MB and biodiesel in general, the direct production of CO_2 from methyl esters is very likely to account for approximately one CO_2 for every two methyl ester molecules consumed, and account for approximately half of the O atoms in the methyl ester fuel. Further experimental and kinetic modeling studies of alkyl ester fuels under both pyrolysis and oxidation conditions are needed to confirm and extend the present analysis.

5. Conclusions

Pyrolysis of methyl butanoate (MB) was studied behind reflected shock waves using laser absorption diagnostics to measure CO, CO_2, CH_3, and C_2H_4 time-histories. The newly developed CO diagnostic using a quantum-cascade laser provides extremely high sensitivity. The experimental results were compared with two detailed kinetic mechanisms and were generally found to be in agreement with the Huynh et al. model. High-temperature measurements show a CO/CO_2 ratio of about 2/1 and CO/MB ratio of very nearly 1/1. These ratios are a combination of the results of the decomposition reactions over the entire pyrolysis history and the later-time results of the H atom abstraction reactions. The current measurements and analysis indicate that very nearly the same proportions of CO and CO_2 production for all linear methyl ester fuels up to and beyond the fuel size for biodiesel should be expected.

Acknowledgments

The experiments were performed at Stanford University. The kinetics work of the Stanford authors was supported by the Combustion Energy Frontier Research Center (CEFRC) and the diagnostic development work by the Air Force Office of Scientific Research (AFOSR) with Drs. J. Tishkoff and C. Li as technical monitors. The work by Dr. Westbrook of Lawrence Livermore National Laboratory was supported by the US Department of Energy with program managers Gurpreet Singh and Kevin Stork.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2012.05.013.

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